

AVA: Research Update and Proposed Direction

Summary for the team · 26 September 2026 · Prepared by Ibrahim
Full list of ~80 verified sources with links: [docs/literature-review.md](#) in the repo. Nothing here is final.
We decide together before the SRS is submitted.

1. The short version

- **Our idea holds up.** Research from 2025–2026 supports every main part of it, and nobody has built it for Pakistan.
- **The model gets simpler.** Drop DuaBiMamba. Start from a model that is already trained to catch deepfakes, cut it to about 7–8 layers, and put a simple classifier on top.
- **“Just fine-tuning” is not a weakness.** The field itself says progress now comes from data, not from new model designs. Our contribution is a new dataset, a live system, two research questions and a human study.
- **Live calls get easier.** Asterisk can quietly tap a call and hand Python plain audio. SIPREC becomes a stretch goal.
- **Pakistani Punjabi data is scarce but reachable**, from Meta's open Shahmukhi set, our own 60–80 speakers, and a free phone-call corpus through a student scholarship.
- **Accents are manageable.** The model listens for AI traces, not accent. We put every accent in both real and fake audio, then measure false alarms per dialect (section 7).
- **Six claims in our current documents are wrong** and must be fixed before Tuesday (section 9).

2. What AVA is, in plain words

A fraudster calls a bank using an AI copy of a customer's voice. The bank's phone system already sends a copy of every call to recording systems, and AVA gets a copy too. Every 2 seconds, AVA takes the last 4 seconds of audio and asks a model: is this a real human or AI? If the answer stays “AI”, an analyst sees an alert. AVA never touches the call; it only listens.

Key idea: the model does not understand words or language. It listens for the tiny sound traces that AI voice generators leave behind. This is why accents and mixed Urdu/English are a data problem, not a coding problem.

3. “Aren't we just fine-tuning someone else's model?”

No. This is the right approach, if we present it correctly.

- Nobody trains big speech models from scratch; it takes thousands of GPU hours. Every winning system in recent deepfake challenges started from a pretrained model.
- The ASVspoof 5 organisers (2026) wrote that new model designs “may be reaching a bottleneck” and that progress now comes from **data design**.
- Microsoft/Nuance (ICASSP 2026): **better data beats bigger models**. Training on realistic phone-call audio improved accuracy on real calls by 57%.

So the model is one component. Our contributions are:

Contribution	Why it is new
Dataset	No Pakistani Punjabi deepfake dataset exists, and no phone-quality Urdu one. Common Voice has no Pakistani Punjabi at all.
Live system	Most research tests saved files. Ours works on live phone calls, on an ordinary CPU, with no internet.
Two research questions	How fast can we be sure a call is fake, and does it wrongly accuse real Punjabi speakers? (Section 4, A and B.)

Contribution	Why it is new
Human study	Pakistani listeners vs AVA on the same clips.

Suggested one-line pitch: “AVA catches AI-cloned voices on live Pakistani phone calls, in Urdu and Punjabi, on an ordinary computer with no internet. We build the first Pakistani phone-call deepfake dataset, and measure how fast and how fairly AVA can decide.”

4. What makes AVA novel

	In simple words	Why a panel will like it	Effort
A	“How fast can we be sure?” AVA listens as the call goes on and alerts only when confident. We measure seconds-to-alert over GSM/AMR phone audio.	Only one unreviewed English paper tried this, not on phone codecs like GSM. A “seconds vs accuracy” graph is easy to show.	Low
B	“Does it wrongly accuse Punjabi speakers?” We measure how often real people of each dialect are flagged as fake.	It turns our accent worry into a finding. Nobody has measured this for Urdu or Punjabi.	Low
C	Humans vs AVA. Play 40 clips to 30–60 people. Who is more accurate: humans, AVA, or humans helped by AVA?	People get only 64–73% right in studies. In the Urdu study, listeners missed about 1 in 3 fakes.	Low
D	“Which seconds are fake?” The dashboard shows a timeline highlighting the fake part of a call.	Real fraud sometimes fakes one sentence. Listeners missed a single fake sentence 77% of the time.	Medium
E	“Is it really the customer?” (stretch goal) Also check the caller against the customer's voice on file.	Matches real bank fraud. It uses a free ready-made model and a simple formula.	Low–med

Recommendation: make A + B the headline research, add C (it is cheap), show D in the dashboard, and keep E as a stretch goal.

5. Technical plan: what changes

Part	Current plan (PRD)	Proposed	Why
Base model	XLS-R 300M	AntiDeepfake MMS-300M , already trained on 74k hours in 100+ languages to catch deepfakes (free)	A better starting point, no extra cost
Classifier head	DuaBiMamba	SLS (simple layer mix + small classifier)	Mamba is only ~1 point better and cannot run on CPU
Layers kept	12	~7–8 , chosen by a quick test	12 is too slow for 8 calls on one CPU
Baseline	AASIST, RawNet2	Add XLS-R + AASIST + RawBoost	The standard baseline, public code
Live calls	Our own SIPREC server	Asterisk “snoop” tap → plain audio to Python. SIPREC via jambonz as a stretch goal.	Asterisk has no SIPREC client; the tap decodes codecs for us
Codecs	G.711, GSM, AMR-NB live	G.711 and GSM live; AMR-NB tested on recorded files	Stock Asterisk cannot decode AMR-NB live
Window	4 s every 2 s	Keep	Shorter windows roughly double errors

Fake-voice generators

Tool	Type	Note
OmniVoice (2026)	Voice cloning	Free; supports Urdu and even Pakistani Punjabi
MMS-TTS	Basic text-to-speech	Free, runs on CPU

Tool	Type	Note
Seed-VC (+ RVC)	Voice conversion	Works in any language. Main source of Punjabi fakes: it keeps the speaker's accent.
Fish Audio S2	Newest “LLM-style” TTS	Held out from training, used only as the unseen-tool test
ElevenLabs v3	Commercial	Only with written permission (its terms ban dataset use)

6. Dataset plan

The dataset is the biggest part of AVA and our main contribution. This section explains what goes into it, how we collect and manage it, and why accents and mixed languages are manageable.

6.1 What the dataset contains

Target: about 50,000 clips of 3–6 seconds each (about 60 hours), roughly 45% real and 55% fake. Each language gets the same share.

	Urdu	Pakistani Punjabi	Total
Real (humans)	~11,000	~11,000	~22,000
Fake (AI), from 4 training generators	~12,000	~12,000	~24,000
Fake from the held-out generator (test only)	~1,500	~1,500	~3,000
Total	~24,500	~24,500	~49,000

Every clip also exists in several **phone-quality versions** (G.711, GSM, AMR-NB, Opus). We count only the original clips towards the 50k, because a codec copy of the same clip is not new data.

6.2 Where the real speech comes from

Real speech	Source	Cost
Urdu (~11k clips)	Common Voice Urdu (80 validated hours, 500+ speakers), capped at 30 clips per speaker so no one dominates; UrduSpeech after a rights check; the Urdu part of our own sessions	Free
Pakistani Punjabi (~11k clips)	Meta Omnilingual pnb_Arab (Shahmukhi, open licence, ~3k clips) + Habib University corpus + our own 60–80 speakers × 10–12 min (~8k clips)	About Rs 60–120k if we pay speakers (estimate)
Other dialects	Common Voice and Meta Omnilingual: Pothwari (14 h), Saraiki (6.75 h), Hindko	Free
Phone-call test set	LDC2017S14: 38.8 h of real Pakistani Punjabi phone calls	Free via LDC Data Scholarship , deadline 15 Jan 2027

6.3 How we record our own speakers

Step	What happens
1. Recruit	60–80 Pakistani Punjabi speakers, about half men and half women, mostly Majhi (Lahore region) plus a few from other areas. Sources: our two Punjabi-speaking members' families and friends, GIKI and Lahore student societies, and one well-connected community contact. (One such promoter raised a Pashto collection drive 108×.)
2. Consent	Everyone signs a consent form before recording, with a separate tick-box for voice cloning. They get an anonymous ID (P001, P002...); real names never go into the dataset.
3. Short form	Age band, gender, home district, dialect, languages spoken, phone model. This lets us report results per dialect later.
4. Session (10–12 min)	Part A, read speech (3 min): 20–30 sentences in Shahmukhi Punjabi and Urdu, some mixed with English. Part B, free speech (5 min): answer simple questions or describe pictures (“tell us about your phone plan”), so people talk naturally and mix languages on their own. Part C, role-play call (3 min): pretend to call a bank about a blocked card. This is exactly the situation AVA protects.

Step	What happens
5. Recording	Over a real phone call into our Asterisk lab (this gives true phone audio), or with a simple web recorder that saves proper WAV files. Not WhatsApp voice notes (see traps).
6. Payment	Rs 1,000–1,500 per session if the budget allows. Volunteers otherwise.

6.4 How we make the fake clips

- **From the same people we recorded.** Every consenting speaker is also used to make fakes, so the same voices appear on both the real and fake side. The model then cannot use “whose voice is this” as a shortcut.
- **Voice conversion (Seed-VC, RVC):** take a real recording and change the voice to someone else's. The words, accent and rhythm stay; only the voice identity changes. This is our main source of Punjabi fakes.
- **Voice cloning (OmniVoice):** give the tool a 10-second sample of a speaker and the same sentences they read, including mixed-language ones.
- **Basic TTS (MMS-TTS):** cheap volume for Urdu.
- **Held-out tool (Fish Audio S2):** used only in the final test, to check AVA on a tool it never saw.
- For every fake we record which tool made it, from which speaker, and from which text.

6.5 Processing pipeline (one script, same for real and fake)

Raw audio → trim silence → cut into 3–6 s clips → match loudness → convert to 16 kHz → make phone versions (G.711, GSM, AMR-NB, Opus, with some packet loss) → automatic quality checks → add a row to the manifest. Real and fake clips go through **exactly the same steps**, so the only difference left between them is “human or AI”.

6.6 How we store and manage it

Topic	Plan
Where audio lives	On the university GPU server or an encrypted drive, with one backup. Never in GitHub: the repo is public, and the .gitignore already blocks audio files.
Manifest	One spreadsheet (CSV) row per clip: clip ID, speaker ID, language, dialect, real/fake, generator, source clip, codec, duration, split, licence. This file is the dataset's “table of contents”.
Scripts	All collection, generation and processing scripts live in the repo's data/ folder, so anyone on the team can rebuild the dataset from scratch.
Versions	ava-data-v0: public data only (Common Voice, Meta, CSALT), early October, so model work can start immediately. ava-data-v1: adds our own recordings and their fakes.
Consent forms	Stored offline and separately, linked only by speaker ID. If someone withdraws, we delete their clips and every fake made from them.
Splits	70% train / 10% dev / 20% test, by person: nobody appears in two splits. Plus separate test sets for the held-out tool, held-out dialects, and real phone calls (LDC).

6.7 How we prove the dataset is good (validation)

- **Labels are correct by construction:** we know who recorded each real clip and which tool made each fake. Nobody has to guess.
- **Automatic checks on every clip:** length, silence, loudness, a noise score, and a language check that it really is Urdu or Punjabi. Fakes also get a speech-recognition check that the words came out right.
- **Manual spot-check:** the team listens to a random 5%. If more than 10% of a batch fails, the whole batch is re-checked (the method used by India's Vaani project).
- **Split check:** a script proves no person appears in both training and testing.
- **Human listening test** (study C), plus a **datasheet** describing sources, consent, speakers, dialects and tools.

6.8 Traps to avoid

- **Make fakes from the same people we recorded,** or the model learns “this voice = real”.
- **Don't use WhatsApp voice notes for real clips only,** or the model learns “compressed = real”.

- **Match silence and length** across real and fake clips. One model “cheated” just by measuring silence.
- **Enough people:** at least 50–100 Punjabi speakers, balanced by gender.

6.9 Rough timeline

When	What
Early October	Ethics approval request; consent form; download public data; build ava-data-v0; test generators on mixed Urdu–English
October – November	Recording sessions with our 60–80 speakers; generate fakes as recordings come in
End of November	ava-data-v1 complete: manifest, splits, quality checks, datasheet
By 15 January 2027	LDC scholarship application for the phone-call test set

7. Why accents and mixed languages are not a big problem

This was the team's biggest worry. The short answer: **we don't need to cover every accent. We need to make sure accent can't be used as a shortcut, and then measure what happens.** Both are straightforward.

Worry	Answer
"Punjabi has too many accents to cover."	The model does not listen to accent or words. It listens for the tiny traces that AI generators leave behind (unnatural timing, phase and texture of the sound). An accent is a property of a human speaker; those traces are a property of the generator. They are different things.
"Won't it learn that some accents are fake?"	Only if accents appear on one side only. Our fakes are made from the same people as our real clips, and voice conversion keeps the original accent. So every accent appears on both sides, and accent carries no information about real vs fake.
"What about accents we never recorded?"	The real risk is a false alarm : an unfamiliar real accent wrongly flagged as fake. We don't guess; we measure it with test-only dialects (Jhangvi, Shahpuri) and report the false-alarm rate per dialect. Nobody has published this for Urdu or Punjabi, so the measurement itself is a contribution (study B).
"We can only record a few people."	Research shows a little data in the target language beats a lot of data in other languages, and about 8 hours of target-language fine-tuning already improves robustness. Our 60–80 speakers plus Meta's Shahmukhi set comfortably pass that.
"Why only Majhi?"	Majhi (the Lahore region) is the basis of standard Punjabi, and it is already our agreed scope. Pothwari and Saraiki are added cheaply from open data, and more dialects are future work.
"People mix Urdu, Punjabi and English."	Same logic: mixed speech goes into real and fake clips in equal amounts (at least 15% each), recorded naturally in Part B of each session and generated from the same mixed sentences. The model can't learn "English = fake". We also report results on mixed clips separately.

In one sentence for the panel: “Accent is not what AVA listens for. We make sure every accent appears in both real and fake audio, and then we measure false alarms per dialect, a measurement nobody has published for Urdu or Punjabi.”

8. Fraud numbers, and which direction AVA protects

Updated figure: NCCIA says **227,757 victims lost Rs 8.32 billion** in 2024–2026. Most cases were WhatsApp hacking, or calls from people pretending to be bank or institution staff (Express Tribune, 11 Sep 2026).

Careful: this number is not directly what AVA stops.

- In those scams, the fraudster calls a citizen and pretends to be bank staff, usually in their **own real voice**. AVA detects AI voices, not lies, so it would not catch a human imposter. AVA also sits at the bank, so it never hears those calls. WhatsApp hacking has no voice at all.
- What the number **does** show: in Pakistan, people and institutions still trust the voice on the phone, and phone impersonation is already the biggest fraud route.
- What cloning changes: today a scammer can only pretend to be a role ("I'm from your bank"). With a 10-second voice sample they can become a specific person. That includes the customer calling the bank to reset a PIN or approve a transfer, which is the direction **AVA protects**.

Evidence for our direction (fraudsters calling into call centres): Pindrop's 2025 report says deepfake attacks on contact centres rose from one every two days (2023) to seven per day (2024), +1,337%. Synthetic-voice fraud at banks rose 149% in 2024. This is US vendor data, so label it as industry data.

The other direction (a cloned voice calling a citizen) could be covered later by a telecom company running AVA on its own network. That is exactly where CTDISR-2025's "keep call data inside Pakistan" rule applies. This is future work, not our scope.

How to say it on a slide: "Phone impersonation is already Pakistan's leading fraud route (Rs 8.32bn, 2024–26). AI voice cloning upgrades it from pretending to be a bank officer to sounding exactly like your customer. AVA protects the institution side of that call."

Other numbers	Status
1,000+ registered call centres; \$328M call-centre exports in FY25	Verified (PSEB / SBP via Business Recorder, ProPakistani)
\$9.3B scam losses, 2.5% of GDP	A survey estimate (GASA/Feedzai 2025); label it as an estimate
Documented Pakistani voice-clone fraud case	None found. Do not claim one.
Industry products (Pindrop, ValidSoft, Resemble...)	None publish results for Urdu or Punjabi. That is our gap.

9. Wrong claims in our current documents (fix before Tuesday)

#	What we wrote	The truth
1	"Coqui XTTSv2 supports Urdu and Punjabi"	It supports neither . Remove it.
2	ElevenLabs as a data generator	Its terms ban using its output to train or test detectors. We need written permission.
3	"AVA complies with CTDISR-2025"	It binds telecom companies , not us. Use it as motivation: telcos must keep call data in-country, so detectors must run on-site.
4	"PDPB 2023" as a law	It is an unpassed draft . Call it "the draft Personal Data Protection Bill".
5	CSALT link csalt.itu.edu.pk	CSALT is at LUMS . Cite the ACL 2024 paper and github.com/CSALT-LUMS.
6	Old briefing: <150 ms, GPU, layers 1–6, 15 months	Outdated. Never quote it.

10. Ethics and approvals

- **We need ethics approval even if we never publish.** Recording and cloning voices counts as human-subjects research, and HEC requires universities to review it. Email GIKI ORIC or the FYP coordinator to ask who approves it.
- **Consent form:** what we record and why; a **separate tick-box for cloning their voice**; storage and deletion date; the right to withdraw; separate tick-boxes for internal use, sharing with examiners, and public release; "voluntary, no effect on grades".
- **Decide now whether we might publish** the dataset or a paper. The consent form must say so from day one.

- Good news: most published deepfake datasets reused audiobooks or scraped celebrity audio **without consent**. Consented recordings put us ahead.
- The GitHub repo is **public**. Never put recordings or consent forms in it.

11. Decisions and to-dos

To do	Owner
Agree on the direction: pitch, novelty A + B + C, simplified technical plan	Everyone
Ask the professor what exactly "50k" means (clips? real only?)	
Email GIKI ORIC about ethics approval	
Email ElevenLabs asking for research permission	
Decide whether we might publish the dataset or a paper	Everyone
Confirm the budget for paying Punjabi speakers (~Rs 60–120k)	Everyone
Assign workstreams: data / model / telephony / app	Everyone
Fix the 6 wrong claims in the slides, then in the PRD/SRS	
Test OmniVoice and Fish Audio with mixed Urdu–English sentences	
Prepare the LDC Data Scholarship application (deadline 15 Jan 2027)	

12. Tuesday: title defense (5 min + 5 min Q&A, everyone speaks)

Speaker	Covers (~75 s each)	Rubric item
1	Problem and motivation: Rs 8.32bn phone impersonation, the cloning upgrade, ethics	Motivation / ethical impact
2	Literature and gap: Urdu detector at 50% error (chance); no Pakistani Punjabi data; no vendor publishes Urdu/Punjabi results	Literature review, problem statement
3	Solution and workflow: call → Asterisk tap → model on CPU → dashboard alert	Engineering knowledge
4	Dataset, novelty (A–C), why it is a complex problem, scope and timeline	Complex-problem attributes

Likely panel questions

- **"You're just fine-tuning a model."** The field says data now matters more than architecture. Our contribution is the dataset, the live system and the new measurements.
- **"Is there enough Punjabi data?"** Meta's open Shahmukhi set, our own 60–80 speakers, and the LDC phone corpus via scholarship. We report the gap honestly.
- **"Can a big model run in real time on a CPU?"** A 2026 study ran a 7-layer version on one CPU core. We keep 7–8 layers and compress to INT8.
- **"How do you test without a real telecom network?"** A free Asterisk phone lab with softphones, and live calls tapped exactly as a bank would.
- **"Ethics?"** Consent with separate cloning opt-in, ethics approval, no public release by default, and AVA only advises (a human decides).

13. Key references

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- 14 NCCIA via Express Tribune, "Pakistanis lose Rs8.32b to cyber fraud", 11 Sep 2026
- 15 Pindrop, 2025 Voice Intelligence and Security Report (industry data)
- 16 PTA, Critical Telecom Data and Infrastructure Security Regulations 2025 (notification of 29 Oct 2025)